

Saliva TwoStep: An RT-LAMP saliva test for SARS-CoV-2 and its assessment in a large population

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Abstract

Here, we develop a simple molecular test for SARS-CoV-2 in saliva based on reverse transcription loop-mediated isothermal amplification (RT-LAMP). The test has two steps: 1) prepare saliva with heat and a stabilization solution, and 2) detect virus by incubating with a primer and enzyme mix at 65°C for 30 minutes. After incubation, saliva samples containing the SARS-CoV-2 genome turn bright yellow, while the negative samples remain pink. Because this test is pH dependent, it can react falsely to naturally acidic human saliva. Here, we report unique buffering conditions and protocols that rendered 295 healthy human saliva samples compatible with the test, producing zero false positives. We also evaluated the test on 168 saliva samples from individuals infected with SARS-CoV-2, and from 47 matched pairs of saliva and anterior nasal samples from infected individuals. The Saliva TwoStep test described here reliably identified infected individuals with 94% sensitivity and >99% specificity.

Introduction

Screening is one of the most basic and powerful tools in the public health arsenal. Screening tests identify underlying illness in apparently healthy or asymptomatic individuals. RT-LAMP is a simple nucleic acid diagnostic concept that has existed for more than 20 years and is used in diverse and even remote settings^{1,2}. RT-LAMP utilizes loop forming primers and strand-displacement polymerases to achieve isothermal amplification of a target RNA template. Therefore, the reaction does not require a thermal cycler, which is a key advantage in keeping screening pipelines from burdening diagnostic pipelines. Successful amplification in RT-LAMP reactions can be directly visualized via various visual indicators, including by phenol red which changes the color in the reaction tube upon successful amplification¹. RT-LAMP assays 1) are specific because they employ four to six primers for each amplification reaction, which in principle reduces off-target effects when compared to PCR, 2) are robust in that they can produce on the order of 10^9 copies of the target in an hour-long reaction¹, and 3) can be performed anywhere because they simply require pipettors and a heating source (e.g. water baths, heat blocks, or thermal cyclers) as equipment. In addition, recent studies have shown that saliva has high diagnostic value for SARS-CoV-2³. Overall, a rapid screening test that combines the simplicity of RT-LAMP and the non-invasive nature of saliva would satisfy the criteria for an effective screening test.

To date, a few studies have attempted to develop RT-LAMP tests to detect SARS-CoV-2 in saliva samples^{4,5}. However, RT-LAMP can have a high rate of false positives^{6,7}. Here, we present an optimized RT-LAMP screening procedure that detects SARS-CoV-2 nucleic acid present in saliva directly, without any need for RNA purification. We optimized buffering and stabilization conditions that neutralize the variability of human saliva and eliminate false

positives. We validated the RT-LAMP test using a large cohort of saliva and matched nasal swab specimens collected from our local university population, comparing the test to two other SARS-CoV-2 tests (one nasal test and one saliva test). We found our optimized RT-LAMP procedure performs consistently with high specificity and sensitivity.

Results

Optimized saliva buffering conditions for RT-LAMP

In colorimetric RT-LAMP, phenol red serves as the visual indicator for nucleic acid amplification. Reactions start as pink/red but turn strongly yellow at pH 6.8 and below. When RT-LAMP amplifies a target, it causes a massive drop in pH within the tube to pH 6.0 – 6.5⁸, triggering the color change to yellow. Human saliva naturally varies in pH between 6.8 and 7.4⁹, posing a significant problem in this pH-dependent assay. In fact, we find about 10% of human saliva samples are naturally acidic enough to immediately trigger phenol red-containing reactions to change to yellow without any target amplification required. If acidic samples are not anticipated and managed, colorimetric RT-LAMP has the potential to produce a high false-positive rate.

To address this issue, we optimized a basic saliva stabilization solution by titrating in various concentrations of sodium hydroxide (NaOH). We performed this optimization using a control RT-LAMP primer set, “RNaseP,” which amplifies the mRNA transcript produced from the human *POP7* gene (primer set developed previously¹⁰). Our goal was to increase the pH of all saliva samples well above the indicator flip-point of pH 6.8, while not making the samples so basic that they couldn’t reach this pH upon successful target amplification. We found that

human saliva containing 14.5 mM NaOH is optimal to inhibit false positives caused by saliva acidity without impeding the intended color change during amplification (**Figure 1**). We designed our saliva stabilization solution to also include a chelating agent (1 mM EDTA final concentration) and Proteinase K to inhibit RNases, both of which help preserve virion RNA and therefore to increase sensitivity. (Note that Proteinase K will inhibit the RT-LAMP reaction if it does not go through a heat inactivation step prior to that reaction.) Finally, the saliva stabilization solution contains TCEP, which aids in RNA stabilization by breaking disulfide bonds present in RNases and helping to reduce saliva viscosity. Our optimized saliva stabilization solution (2X solution: 5 mM TCEP, 2 mM EDTA, 29 mM NaOH, 100 µg/mL Proteinase K, diluted in DEPC-treated water) is key to this test and we find that it rendered all saliva samples tested (n=463) robust to colorimetric RT-LAMP. For additional advice on controlling the acidity of reactions, which is key to reducing false positives in RT-LAMP, see **Supplemental Text S1**.

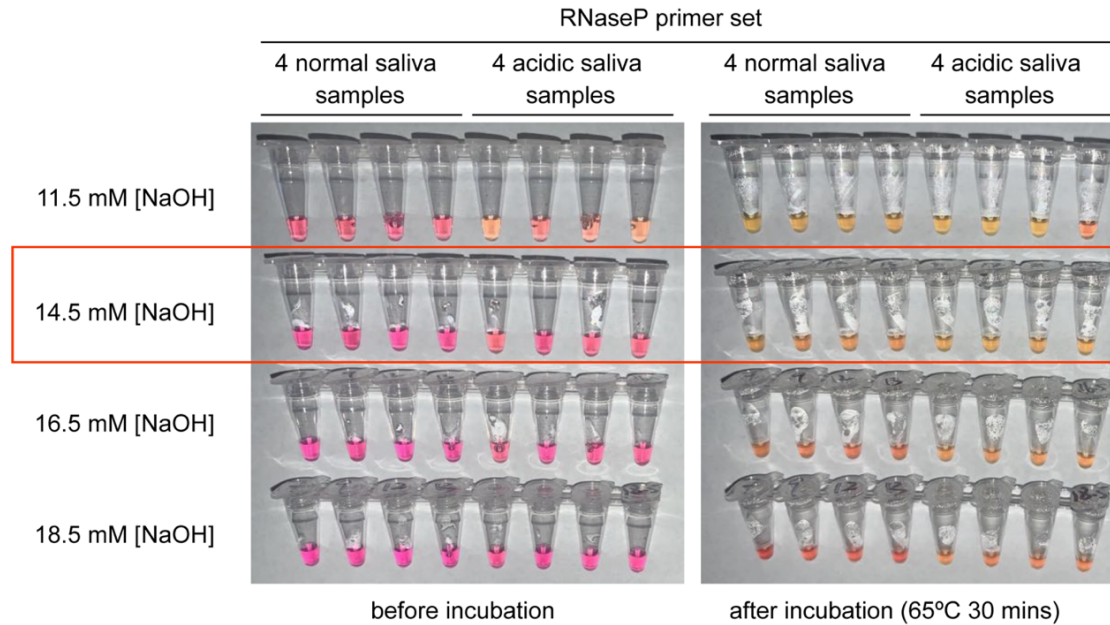


Figure 1: Optimized saliva pH for input into RT-LAMP. Naturally variable saliva samples were mixed 1:1 with 2X saliva stabilization solution containing NaOH at various concentrations (final molarity of NaOH after mixing shown). Samples were then heated at 95°C for 10 minutes, and then analyzed by RT-LAMP using a primer set targeting the human RNaseP transcript. Tubes that turn yellow have a successful amplification of this RNA. The red box indicates the final optimal concentration chosen. In this row, none of the samples are yellow before heat incubation for isothermal RT-LAMP amplification, but all samples are yellow after incubation.

Optimized RT-LAMP primer sets for detecting SARS-CoV-2 in human saliva

A critical parameter in RT-LAMP is primer design because RT-LAMP requires 4-6 primers all working together¹. We found that the “AS1E” set, developed by Rabe et. al. and targeting the *ORF1ab* region of the SARS-CoV-2 genome, performs well⁵. However, in order to target two distinct regions from the SARS-CoV-2 genome, we designed and tested a large number of additional primer sets. Two of our custom sets, “ORF1e” targeting the virus *ORF1ab* gene, and “N2” targeting the virus *N* gene, exhibited similar sensitivity and amplification efficiency as the AS1E set, as determined using real-time fluorescence monitoring of RT-LAMP

products (**Figure 2A**). We next confirmed that these primer sets were both compatible with saliva preserved in our saliva stabilization solution and with colorimetric RT-LAMP (**Figure 2B**).

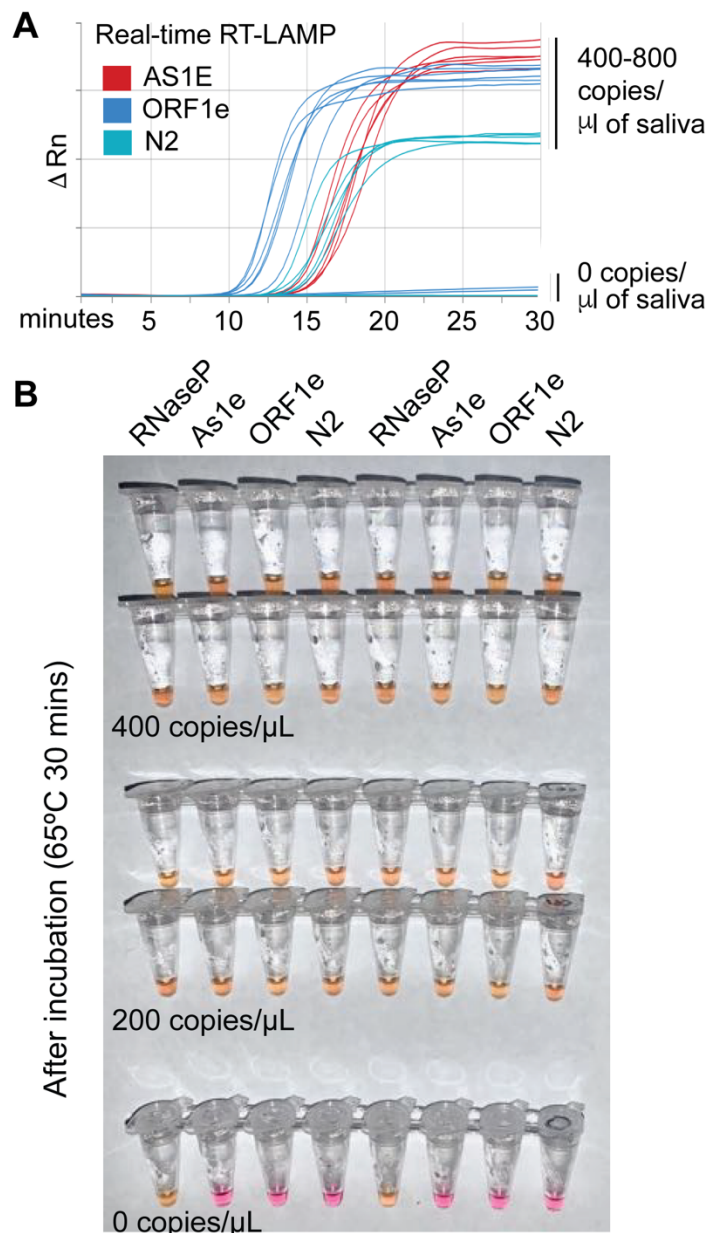


Figure 2: Optimized RT-LAMP primer sets for detecting SARS-CoV-2 in human saliva. A) Three RT-LAMP primer sets targeting the SARS-CoV-2 genome (AS1E, ORF1e, and N2) were tested with real-time RT-LAMP. Saliva was mixed 1:1 with 2X saliva stabilization solution, heated at 95°C for 10 minutes, and then *in vitro* transcribed SARS-CoV-2 RNA was spiked in at the indicated concentrations. 4 μ l was added to a master mix containing primers and NEB's WarmStart LAMP 2x Master Mix in a final reaction volume of 20 μ l. Reactions were incubated at 65°C and a fluorescence reading was taken every 30 seconds. EvaGreen was used to monitor amplification products in real-time (X-axis) using a QuantStudio3 quantitative PCR machine. There are 9 lines for each of the three primer sets because three concentrations of spiked in SARS-CoV-2 RNA were each tested in triplicate (0, 400, 800 copies / μ l saliva). The normalized change in fluorescence signal (ΔRn) is shown on the Y-axis. **B)** Saliva mixed 1:1 with 2X saliva stabilization solution was heated (95°C for 10 minutes) and then spiked with SARS-CoV-2 RNA at the indicated concentrations. Replicates were tested by RT-LAMP with the control RNaseP primer set and three distinct SARS-CoV-2 primer sets (AS1E, ORF1e, and N2).

Addressing biosafety concerns of sample handling

Next, we addressed the biosafety concerns of handling samples. Recent studies suggest that incubation for 3 minutes at 95°C is sufficient to inactivate SARS-CoV-2 virions¹¹. However, when heating saliva samples for downstream analysis of RNA, one must balance heating long enough to liberate the target RNA from virions with not heating for so long that the target RNA will be degraded. Heating at 95 °C does degrade SARS-2-CoV RNA that is spiked directly into saliva samples but does not degrade viral RNA when it is spiked into samples within SARS-CoV-2 virions (**Supplemental Figure S1**). A 10-minute incubation of saliva samples at 95°C was found to be optimal (**Supplemental Figure S1**). We designed our test procedure such that testing personnel avoid handling open tubes until after this step to increase biosafety.

Assessment of sample stability

Stability of saliva samples from the time of collection to the time of processing and analysis is important if testing cannot be performed immediately, or if the tests are being conducted in batches. Saliva samples containing purified virions and diluted with 2X saliva stabilization solution were stored at 4°C for 24, 48, or 96 hours before being inactivated at 95°C and analyzed using colorimetric RT-LAMP (**Supplemental Figure S2**). We tested saliva collection and storage over a range of SARS-CoV-2 virion spike-in concentrations. We observed no significant changes in sample stability over this time course, suggesting that saliva samples stored in saliva stabilization solution at 4°C are stable for at least four days.

Determining the limit of detection

We next sought to carefully evaluate the limit of detection for this test. The lowest concentration at which positive samples were reliably identified was 200 virions/ μ L in saliva (red box, summary table in **Figure 3A**). We next tested 20 replicates at this concentration (200 virions/ μ L) using all four primer sets (**Figure 3B**). The ORF1e primer set was not consistent in its performance at 200 virions/ μ L. Therefore, we decided to eliminate the ORF1e primer set from our testing panel and define a final colorimetric RT-LAMP test that includes primer sets RNaseP, AS1E, and N2. Note that the limit of detection refers to the virus concentration that can be identified > 95% of the time, and the assay does sometimes detect the virus at even lower concentrations.

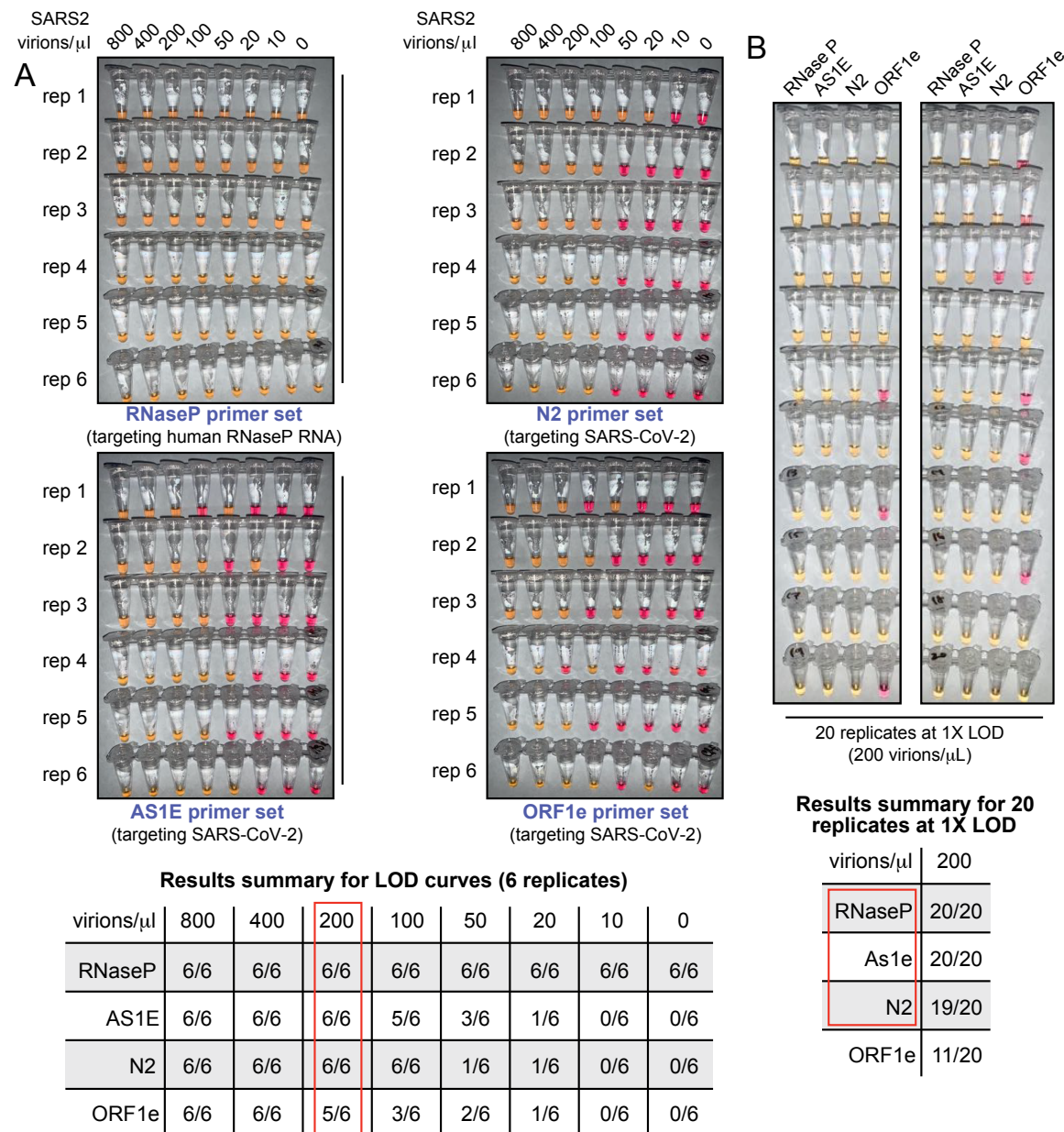


Figure 3: SARS-CoV-2 virion limit of detection using RT-LAMP and saliva samples. A) Saliva samples were spiked with the indicated concentrations of heat-inactivated SARS-CoV-2 virions (top) before being diluted 1:1 with 2X saliva stabilization solution. Samples were then heated at 95°C for 10 minutes and subjected to RT-LAMP at 65°C for 30 minutes in 6 replicates. Each panel represents a unique primer set (listed at the bottom of each panel). The table at the bottom shows a summary of positive reactions observed (yellow). Red box indicates the determined RT-LAMP limit of detection (LOD). **B)** Saliva samples were spiked with heat-inactivated SARS-CoV-2 virions at a concentration of 200 virions/ μ L (the limit of detection of our assay) before being diluted 1:1 with 2X saliva stabilization solution. Samples were then heated at 95°C for 10 minutes and 20 replicates of RT-LAMP with the indicated primer sets were incubated at 65°C for 30 minutes. The table at the bottom shows a summary of positive reactions (yellow). Red box indicates our selection of primer sets to advance to subsequent analysis.

We next performed a blinded study. Heat-inactivated virions were spiked into human saliva at various concentrations at or above the limit of detection, and these as well as uninfected saliva samples were blinded and passed to a second member of our personnel. After running the RT-LAMP test on 60 such samples, only one positive sample did not score correctly. In that sample the SARS-CoV-2 primer set N2 failed (**Supplemental Figure S3**). All negative samples were scored correctly (100% specificity, binomial 95% confidence interval [88%,100%]). Conservatively counting the inconclusive test as a false negative lead to a sensitivity estimate of 97% (binomial 95% confidence interval [93%,100%]). See **Figure 4B** for a breakdown by primer set.

Evaluation on human samples

SARS-CoV-2 screening was initiated on the University of Colorado Boulder campus starting in the summer/fall of 2020. Saliva samples were taken weekly from residents of dormitories and at several testing sites throughout the campus. Participants were asked to refrain from eating or drinking 30 minutes prior to sample collection, though we expect this may not have been universally followed. All saliva samples were first analyzed by a published quantitative RT-PCR method ¹², and so each sample is matched with a Ct value from this test. During campus testing, 295 negative samples, and 168 positive samples with viral loads at or above our limit of detection (200 virions/ μ L) were also evaluated with the Saliva TwoStep test described here. These samples had already been heat inactivated for 30 minutes at 95°C as the initial step of the quantitative RT-PCR protocol. Since the heating component of our Saliva Preparation step had already been performed, an aliquot of the sample was transferred into our 2X saliva stabilization solution (without Proteinase K) and then put through the RT-LAMP reaction as described above. For each of the 463 samples, three reactions were performed with

different primer sets: RNaseP (positive control), AS1E, and N2 primer sets (the latter two sets detecting SARS-CoV-2). During this part of the study, we noticed that decreasing the input sample amount from 4 μ L to 2 μ L in a total reaction volume of 20 μ L further increases tolerance of the RT-LAMP reaction color to acidic saliva samples because less saliva is added. We thus reduced the input sample amount to 2 μ L when evaluating these human samples. For all 463 samples, RT-LAMP with primers to human RNA positive control (RNaseP) turned positive (yellow).

Specificity: 295 saliva samples tested negative for SARS-CoV-2 by quantitative RT-PCR were used for evaluation. We re-tested all of those samples with Saliva TwoStep to evaluate our false-positive rate. Remarkably, for all 295 SARS-CoV-2-negative samples, AS1E and N2 primers sets both universally returned a result of negative, consistent with the quantitative RT-PCR results. Therefore, there were zero false positives and the test has a specificity of 100% in this extensive sample set. This shows the strength of our saliva stabilization solution, which mitigates the problem of false-positives in RT-LAMP due to some human saliva samples being naturally acidic.

Sensitivity: We next analyzed 168 SARS-CoV-2-positive samples with viral loads above the limit of detection of our assay. We determined the relative viral load of each positive saliva samples based on the reported Ct values and the quantitative RT-PCR standard curve (**Supplemental Figure S4**). Of these samples, 158 (94%; AS1E primers) or 142 (85%; N2 primers) returned positive RT-LAMP test results (**Figure 4A**). In **Figure 4B**, we summarize the performance of both primer sets in both this test of human saliva samples and in the spiked in virion experiments described above (**Supplemental Figure S3**). The observed limit of detection of the AS1E primer set was determined from this data to be 266 virions/microliter. The strong

congruence with our prior estimate of 200 virions/microliter demonstrates that heating for 30 minutes prior to adding stabilization solution and using 2 μ L of saliva instead of 4 μ L both have very little effect. Because the AS1E primer set performs best throughout our study, we include that as the main primer set in the Saliva TwoStep test. However, the N2 primer set still performs well and can be used when it is desirable to detect a second region of the SARS-CoV-2 genome.

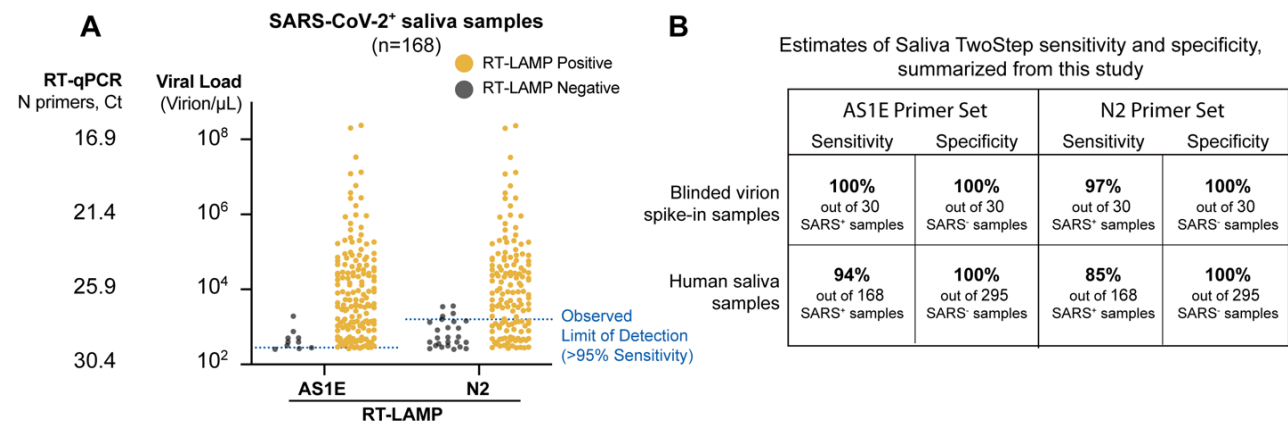


Figure 4. Evaluation of the performance of RT-LAMP on SARS-CoV-2 positive human samples. **A)** Human saliva samples were analyzed for SARS-CoV-2 using quantitative RT-PCR method. The remaining saliva was mixed 1:1 with 2X saliva stabilization solution and re-tested using RT-LAMP. The SARS-CoV-2 positive saliva samples that have viral load above the RT-LAMP limit of detection (200 virions/ μ L) were shown. The results of RT-LAMP are compared to relative saliva viral load determined by quantitative RT-PCR. The figure shows the distribution of the viral load of all 168 positive saliva samples separated by the corresponding RT-LAMP reaction results with either the AS1E or N2 primer-set. **B)** A summary of the sensitivity and specificity of the Saliva TwoStep test, from both the data in panel A, and from the blinded sample evaluation described above and shown in **Supplemental Figure S2**.

Test sensitivity as a function of viral load in the sample: For both primer sets, we calculated the sensitivity (positive agreement with quantitative RT-PCR) and specificity (negative agreement with quantitative RT-PCR) of the RT-LAMP test at various levels of viral load cutoffs (**Table 1**). The differences in the observed limit of detection between the two SARS-CoV-2 primer sets could reflect the differences in the primer efficiencies, as well as the dynamics in relative viral transcript abundance¹³.

Table 1. Summary of RT-LAMP evaluation in human samples

			RT-LAMP				
			No. of Samples	AS1E		N2	
				No. of Positives	Agreement	No. of Positives	Agreement
Quantitative RT-PCR (SARS-CoV-2 N)	Negative		295	0	295/295 (100%)	0	295/295 (100%)
	Positive (Levels of Viral Load: Virions/ μ L)	4000	82	82	82/82 (100%)	82	82/82 (100%)
		2000	97	97	97/97 (100%)	94	94/97 (96.9%)
		1000	118	117	117/118 (99.2%)	110	110/118 (93.2%)
		800	123	122	122/123 (99.2%)	112	112/123 (91.1%)
		400	143	139	139/143 (97.2%)	129	145/173 (90.2%)
		200	168	158	158/168 (94.0%)	142	142/168 (84.5%)

Assessment of Saliva TwoStep against an EUA approved nasal swab test.

For 169 of the SARS-CoV-2-positive saliva samples analyzed above, 47 also had matched nasal samples collected no more than two days later. We next compared the results of the Saliva TwoStep test with the result obtained by the Quidel Lrya nasal swab RT-PCR test. The RT-LAMP produced one false negative in this sample set, whereas the Lyra test produced three (**Figure 5A**). However, this is still remarkably consistent given that this comparison involves three degrees of freedom: biosample (saliva versus nasal swab), test modality (RT-PCR versus RT-LAMP), and days between saliva and nasal samples collection (up to two days

apart). A summary of how these first two degrees of freedom affect test congruency are shown in **Figure 5B**.

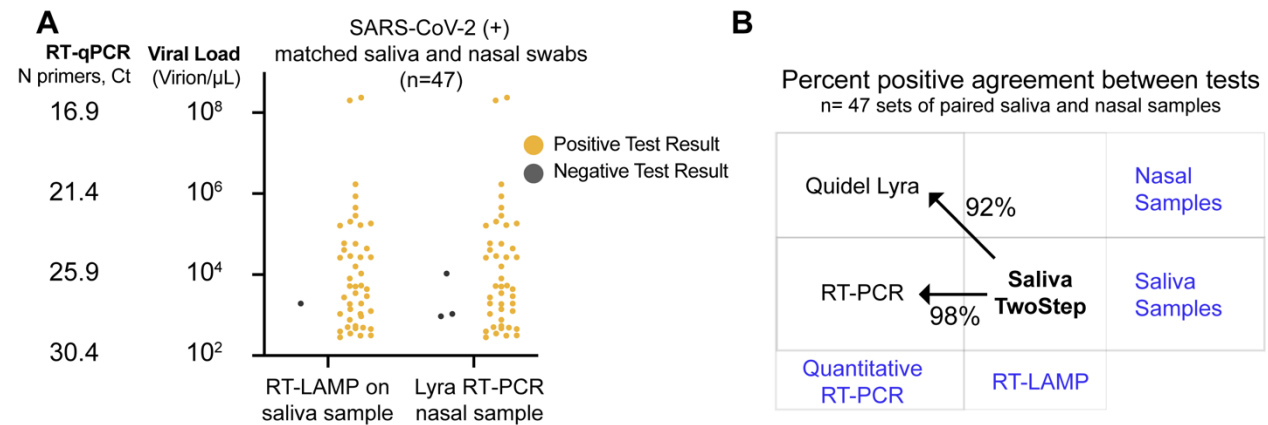


Figure 5. Assessment of Saliva TwoStep against an EUA approved nasal swab test.

(A) Matched nasal swabs and saliva from 47 individuals were analyzed (all of whom were SARS-CoV-2 positive at the time that these samples were collected, as verified by an original test). Nasal swab samples were tested using the EUA Quidel Lyra RT-PCR test. The saliva samples from those same individuals were tested with the Saliva TwoStep test. Data points represent individuals ($n = 47$), and the corresponding test result is color-coded: positive, yellow; negative, grey. **(B)** Positive test agreement between Saliva TwoStep and the two comparator tests. The nature of the sample used by each test (nasal swab or saliva), and the test chemistry (quantitative RT-PCR or RT-LAMP) are delineated.

Final test conditions

From the experiments described above we selected the final optimized conditions for our Saliva TwoStep test. The two steps have an end-to-end processing and analysis time of approximately 45 minutes (**Figure 6**).

Step 1. Prepare Saliva: Collect saliva, combine 1:1 with 2X saliva stabilization solution and incubate at 95°C for 10 minutes. *Note:* We have determined that performing a heating step at 95°C for 30 minutes before addition of the saliva stabilization solution also works equally well. However, in this case Proteinase K must be omitted.

Step 2. Detect Virus: Incubate at 65°C for 30 minutes: 2 µL saliva from step 1, 10 µL 2X NEB RT-LAMP enzyme mix, 6 µL of nuclease-free water and 2 µL 10x primer mix for a final reaction volume of 20 µL.

Step 3. Reaction Inactivation (optional): Stop reaction at 80°C for 2 minutes. This stabilizes color so that results can be analyzed at a later time. The multiple heating steps here may be programmed into a thermal cycler for maximum convenience, but this is not necessary.

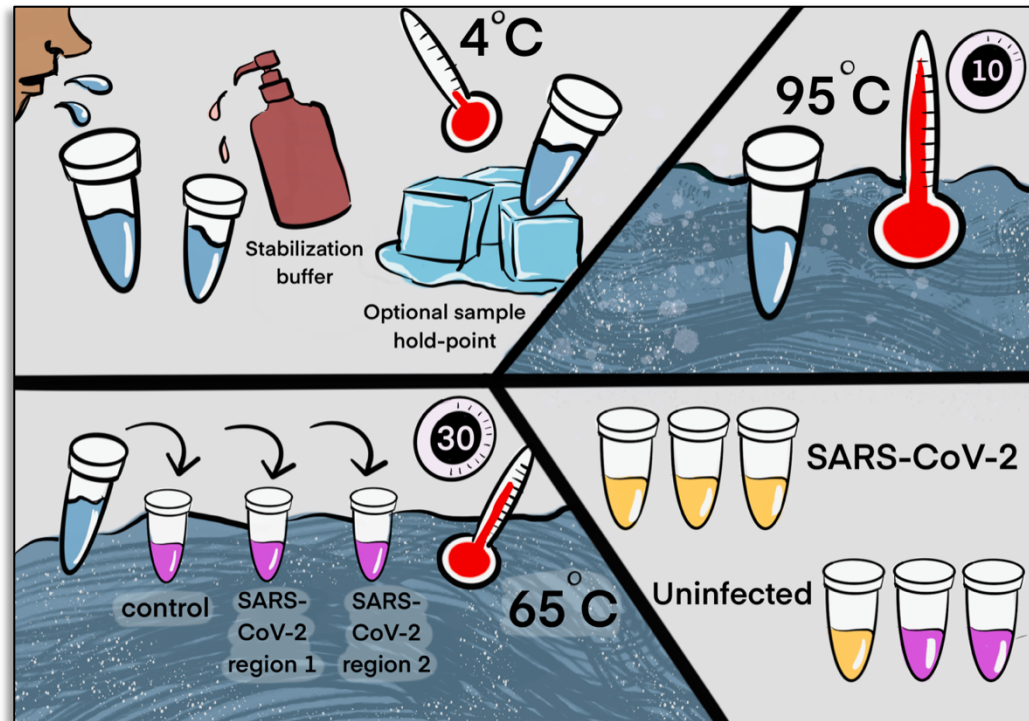


Figure 6. Two step detection of SARS-CoV-2 in saliva.

Upper half) Step 1: Prepare Saliva. Person provides 1 mL of saliva, and 1 mL of 2X saliva stabilization solution is then added to it. (This sample can be processed immediately or stored in the refrigerator at 4°C for at least 4 days.) The mixture is heated at 95°C for 10 minutes. This step serves to neutralize the pH of saliva, liberate viral RNA from virions in the saliva, and inactivate virions for safe handling¹¹ (although appropriate safety precautions should always be taken). We have determined that performing a heating step at 95°C for 30 minutes before addition of the saliva stabilization solution also works equally well. However, in this case Proteinase K must be left out of that solution.

Lower half) Step 2: Detect Virus. 2 µL of stabilized saliva from step 1 is pipetted into each of three test tubes pre-filled with the RT-LAMP master mix and primers. The only thing different between the three tubes is the primer set included, with each set targeting either the human positive control RNA or a region of SARS-CoV-2 RNA, as indicated. After incubation, the reaction will turn from pink to yellow if the target RNA is present in saliva. An example of a positive and a negative test are shown.

Graphic by Annika Rollock.

Discussion

There are several advantages to the SARS-CoV-2 Saliva TwoStep RT-LAMP screening approach described here: **1)** The use of saliva sample eliminates invasive nasal swab-based

sampling, which requires special supplies and causes discomfort. **2)** We optimized saliva stabilization solution that allows for the neutralization of a broad range of naturally acidic saliva samples while maintaining compatibility with a colorimetric RT-LAMP assay. We did not obtain a single false positive. The solution also helps preserve saliva samples for at least four days before processing. **3)** We determined the optimal sample heating condition that liberates the host and viral RNA with minimal degradation. The simple heating step increases biosafety and avoids formal RNA extraction procedures. **4)** For RT-LAMP, we incorporated additional primers based on up-to-date SARS-CoV-2 genome databases and identified primers allowing efficient target amplification. **5)** Overall, with the simplified two steps of saliva preparation and virus detection, the test has a rapid sample-to-result turnaround time of 45 minutes. Through the optimization process, we realized the potential sources of false positive results and provided a detailed summary for troubleshooting (**Methods and Supplemental Text S1**). By strictly following these application notes, we completely avoided false positive results during the evaluation of a large cohort of human saliva samples, achieving 100% specificity. We also evaluated the RT-LAMP test performance based on the experimentally-determined limit of detection. Using SARS-CoV-2 positive human saliva samples, we confirmed that the RT-LAMP test can consistently identify infected individuals with 94% sensitivity.

Based on these observations, we conclude that the Saliva TwoStep described herein can be used as a SARS-CoV-2 screening tool to reliably identify highly infective individuals with minimal laboratory setup, potentially serving as a tool for effective SARS-CoV-2 surveillance at the community level. This RT-LAMP testing offers many solutions to a nation-wide shortage of COVID-19 testing. Combined with the protocol presented here that enables the use of saliva samples, this rapid community testing would provide more frequent tests to those who face the risk of transmission every day, but who would not have the time or resources to seek tests in a

clinical setting. With minimal set-up this test could be performed in diverse settings such as factories, office buildings, or schools.

Ethics statements

This study was approved by the University of Colorado Boulder Institutional Review Board.

Saliva samples for assay development were collected under protocol 20-0068. Testing on human subjects is aggregated data resulting from University of Colorado Boulder operational COVID-19 surveillance testing.

Conflict of interest statement

Some of the authors of this study (NRM, QY, CLP, SLS) are founders of Darwin Biosciences, who licenses the RT-LAMP assay described herein.

Acknowledgements

We dedicate this study to the memory of our wonderful colleague, Denise Muhlrud. We thank Dan Larremore for invaluable advice, and the University of Colorado Boulder for their ambitious and aggressive approach to keeping our campus safe. In particular, we thank Kristen Bjorkman for playing a key part in making it all happen.

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Material and Methods

RT-LAMP primer design and preparation

Regions of the SARS-CoV-2 genome that are conserved among strains were identified using genome diversity data from NextStrain (nextstrain.org/ncov/global). Next, nucleotide-BLAST (blast.ncbi.nlm.nih.gov) was used to filter out genome sequences that share high sequence homology with other seasonal coronavirus genomes. Finally, PrimerExplorer V5 (primerexplorer.jp/e/) was used to design RT-LAMP primers targeting the specific regions of SARS-CoV-2 genomes. The F3, B3, FIP, BIP, Loop F and Loop B primers were selected for optimal melting temperature and complementarity using A plasmid editor (ApE). All primers were ordered from IDT in desalted form. In all cases, a 10X concentration of primer sets was made containing 16 μ M FIP and BIP primers, 4 μ M LF and LB primers, and 2 μ M F3 and B3 primers.

All primers should be ordered with HPLC purification, which ensures the yield and avoids cross contamination from other SARS-CoV-2-related synthesis projects being run on the same equipment at the oligo synthesis facilities (which can lead to false positives). This is particularly a problem during a pandemic where these facilities are handling many oligo synthesis orders focused on the same pathogen¹⁴. It is also recommended that you communicate with the primer synthesis company to inform them that primers are intended for use with a SARS-CoV-2 screening test. Several companies have dedicated facilities for minimizing cross-contamination of SARS-CoV-2 templates. In addition, primers should be diluted in nucleus-free water, instead of Tris-EDTA (TE) buffer, which will also inhibit pH change that takes place during RT-LAMP.

Primer Set Name	LAMP Primer Component	Primer Sequence (5' - 3')
“RNaseP” amplifies human RNA for positive control ¹⁰	F3	TTGATGAGCTGGAGCCA
	B3	CACCCTCAATGCAGAGTC
	Loop F	ATGTGGATGGCTGAGTTGTT
	Loop B	CATGCTGAGTACTGGACCTC
	FIP	GTGTGACCCTGAAGACTCGGTTTTAGCCACTGACTCGGATC
	BIP	CCTCCGTGATATGGCTCTTCGTTTTTTCTTACATGGCTCTGGTC
“AS1E”	F3	CGGTGGACAAATTGTCAC
	B3	CTTCTCTGGATTAAACACACTT
	Loop F	TTACAAGCTTAAAGAATGTCTGAACACT
	Loop B	TTGAATTTAGGTGAAACATTTGTCACG
	FIP	TCAGCACACAAAGCCAAAAATTTATTTTTCTGTGCAAAGGAAATTAAGGAG
	BIP	TATTGGTGGAGCTAACTTAAAGCCTTTTCTGTACAATCCCTTTGAGTG
“N2” developed herein	F3	CGGCAGTCAAGCCTCTTC
	B3	TTGCTCTCAAGCTGGTTCAA
	Loop F	<i>This set does not require a Loop F primer</i>
	Loop B	ATGGCGGTGATGCTGCTCTT
	FIP	TCCCCTACTGCTGCCTGGAGCGTTCCTCATCACGTAGTCG
	BIP	TCTCCTGCTAGAATGGCTGGCATCTGTCAAGCAGCAGCAAAG
“ORF1e” developed herein	F3	GGCTAACTAACATCTTTGGC
	B3	GTCAGCACACAAAGCCAA
	Loop F	TCTTCAAGCCAATCAAGGAC
	Loop B	TTGTCGGTGGACAAATTGT
	FIP	TCTCTAAGAACTCTACACCTTCCTTTTACTGTTTATGAAAACTCAAACC
	BIP	TATCTCAACCTGTGCTTGTGAAATTTTAGAATGTCTGAACACTCTCCT

SARS-CoV-2 RNA and virion standards

Synthetic SARS-CoV-2 RNA control (Twist Bioscience #102019) was obtained and its copy number of 1×10^6 copies/ μ l was confirmed using RT-qPCR in conjunction with a DNA plasmid control containing a region of the N gene from the SARS-CoV-2 genome (IDT #10006625).

Heat-inactivated SARS-CoV-2 virion control (ATCC #VR-1986HK) was obtained and its

concentration of 3.75×10^5 virions/ μL was confirmed using RT-qPCR in conjunction with both the synthetic SARS-CoV-2 RNA control and a DNA plasmid control containing a region of the N gene from the SARS-CoV-2 genome. SARS-CoV-2 RNA was added to saliva samples after being mixed 1:1 with saliva stabilization solution and heated at 95°C for 10 minutes, unless stated otherwise, whereas heat-inactivated SARS-CoV-2 virions were added to saliva samples and mixed 1:1 with saliva stabilization solution before being heated. Concentrations reported throughout this study represent the final concentration of standards in saliva before it was mixed 1:1 with 2X saliva stabilization solution.

Saliva preparation with heat and stabilization solution

When making the 2X saliva stabilization solution, we offer several key pointers: 1) Use TCEP-HCl (GoldBio #TCEP10). The -HCL form must be used to produce the correct final stock pH. 2) Use EDTA, 0.5 M, pH 8.0 (Sigma-Aldrich #324506). It is important to use a pH 8.0 stock solution, otherwise this also affect the pH of the final stabilization solution. 3) Use lyophilized Proteinase K (Roche # 3115879001). It is important to use the lyophilized form. Liquid forms will contain Tris, which inhibits the pH change during the RT-LAMP reaction. 4) 10 M NaOH was prepared by dissolving NaOH pellets (Sigma-Aldrich #221465) into nuclease-free water, before being added to the 2X solution to reach the correct concentration. The following is the exact recipe that we used to create a 100mL stock of 2X saliva stabilization solution:

Components	Amount Mixed	Final Concentration in 2X stock solution
TCEP-HCl (GoldBio #TCEP10)	143.3 mg	5 mM
0.5 M EDTA, pH 8.0 (Sigma-Aldrich #324506)	400 μL	2 mM
10 M NaOH (Sigma-Aldrich #221465)	290 μL	29 mM
Proteinase K (Roche #3115879001)	10 mg	100 $\mu\text{g/mL}$

Nuclease-free water	To 100 mL	
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Saliva samples (1 mL) were collected in sterile, nuclease-free 5 mL conical screw-cap tubes (TLD Five-O # TLDC2540). 2X saliva stabilization solution described above was then added at a 1:1 ratio. Samples were shaken vigorously for 5 -10 seconds and incubated at 95°C for 10 minutes. Samples were then placed on ice before being used in downstream analyses.

Real-time RT-LAMP

For each reaction, 10 µL WarmStart LAMP 2X Master Mix (NEB #E1700) was combined with 1 µL 20X EvaGreen Dye (Biotium #31000), 2 µL 10X primer mix, and 3 µL DEPC-treated water. The combined reaction mix was added to MicroAmp Optical 96-Well Reaction Plate (ThermoFisher #N8010560) and then 4 µL processed saliva sample was added. The reaction was mixed using a multi-channel pipette and incubated in Applied Biosystems QuantStudio3 Real-time PCR system. The reaction proceeded at 65°C for 30 minutes with fluorescent signal being captured every 30 seconds. The results were visualized and analyzed using ThermoFisher's Design and Analysis software.

Colorimetric RT-LAMP

WarmStart Colorimetric LAMP 2X Master Mix (NEB #M1800) was used in all colorimetric RT-LAMP reactions. Each reaction was carried out in a total of 20 µL (10 µL WarmStart Master Mix, 2 µL 10X primer mix, 4 µL processed saliva sample, and 4 µL DEPC-treated water). Reactions were set up in PCR strip tubes on ice. Saliva template was added last and tubes were inverted several times to mix samples and briefly spun down in a microfuge. Reactions were incubated in

a thermal cycler at 65°C for 30 minutes and then deactivated at 80°C for 2 minutes. The incubation was carried out without the heated lid to simulate a less complex heating device. Images of reactions were taken using a smartphone. For the community deployment of this assay, 2 µl of processed saliva was used instead of 4 µl.

Testing of University Samples

For sample collection conducted at our university, individuals are asked to collect no less than 0.5 mL of saliva into a 5 mL screw-top collection tube. Saliva samples were heated at 95°C for 30 minutes to inactivate the viral particles for safe handling, and then placed on ice or at 4°C.

For quantitative RT-PCR analysis, the university testing team transferred 75 µL of saliva into a 96-well plate where each well had been pre-loaded with 75 µL TBE buffer supplemented with 0.5% Tween 20. (The remaining saliva in the 5 mL collection tube proceeded to RT-LAMP testing as described in the next paragraph.) 5 µL of the diluted sample was then added to a separate 96-well plate containing 15 µL reaction mix composed of: TaqPath 1-step Multiplex Master Mix (Thermo Fisher A28523), triplex primer mix consisting of: nCOV_N2 forward and reverse primers from IDT (10006833 and 10006834) at 600 nM final concentration, nCOV_N2 (SUN) Probe (10007049) at 300 nM final concentration, E_Sarbeco_F1 forward and E_Sarbeco_R2 reverse primers (10006888 and 10006890) at 400 nM final concentration, E_Sarbeco_P1 (FAM) Probe (10006893) at 200 nM final concentration, RNaseP forward and reverse primers (10006836 and 10006837) at 50 nM final concentration and RNaseP (ATTO™ 647) Probe (10007062) also at 50 nM final concentration, and nuclease-free water. The reactions were mixed, spun down, and loaded onto a Bio-Rad CFX96 or CFX384 qPCR machine. Quantitative RT-PCR was run using the standard mode, consisting of a hold stage

(25°C for 2 minutes, 50°C for 15 minutes, and 95°C for 2 minutes) followed by 44 cycles of a PCR stage (95°C for 3 seconds, 55°C for 30 seconds, with a 1.6°C/sec ramp up and ramp down rate). In the study herein, Ct values from the nCOV_N2 primer set are used.

Leftover samples from this testing procedure were then tested with RT-LAMP. 50 µl of saliva samples was transferred and mixed into a 96-well plate, pre-loaded with 50 µl 2X saliva stabilization solution without proteinase K (5 mM TCEP-HCL, 2 mM EDTA, 29 mM NaOH, diluted in DEPC-treated water). 2 µL of diluted saliva samples were transferred into 8-strip PCR tubes containing RT-LAMP reaction mixture (enzymes and primers). For each sample, three RT-LAMP reactions were carried out to amplify human RNaseP as a control and AS1E and N2 for SARS-CoV-2. The reactions were incubated at 65°C for 30 minutes followed by inactivation at 80°C for 2 minutes on a thermal cycler (Bio-RAD T100). A color change from pink to yellow was observed visually to interpret results.

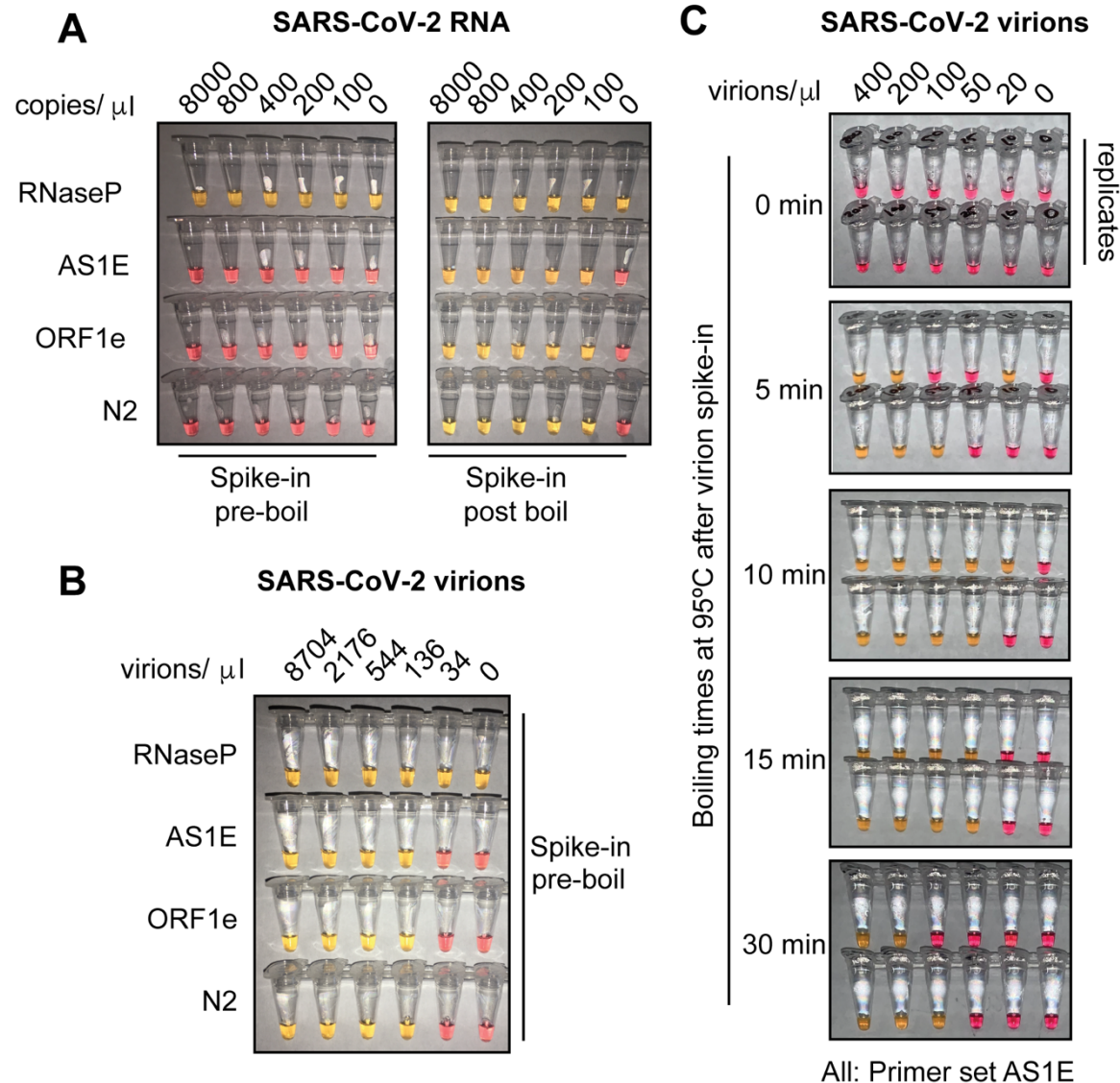
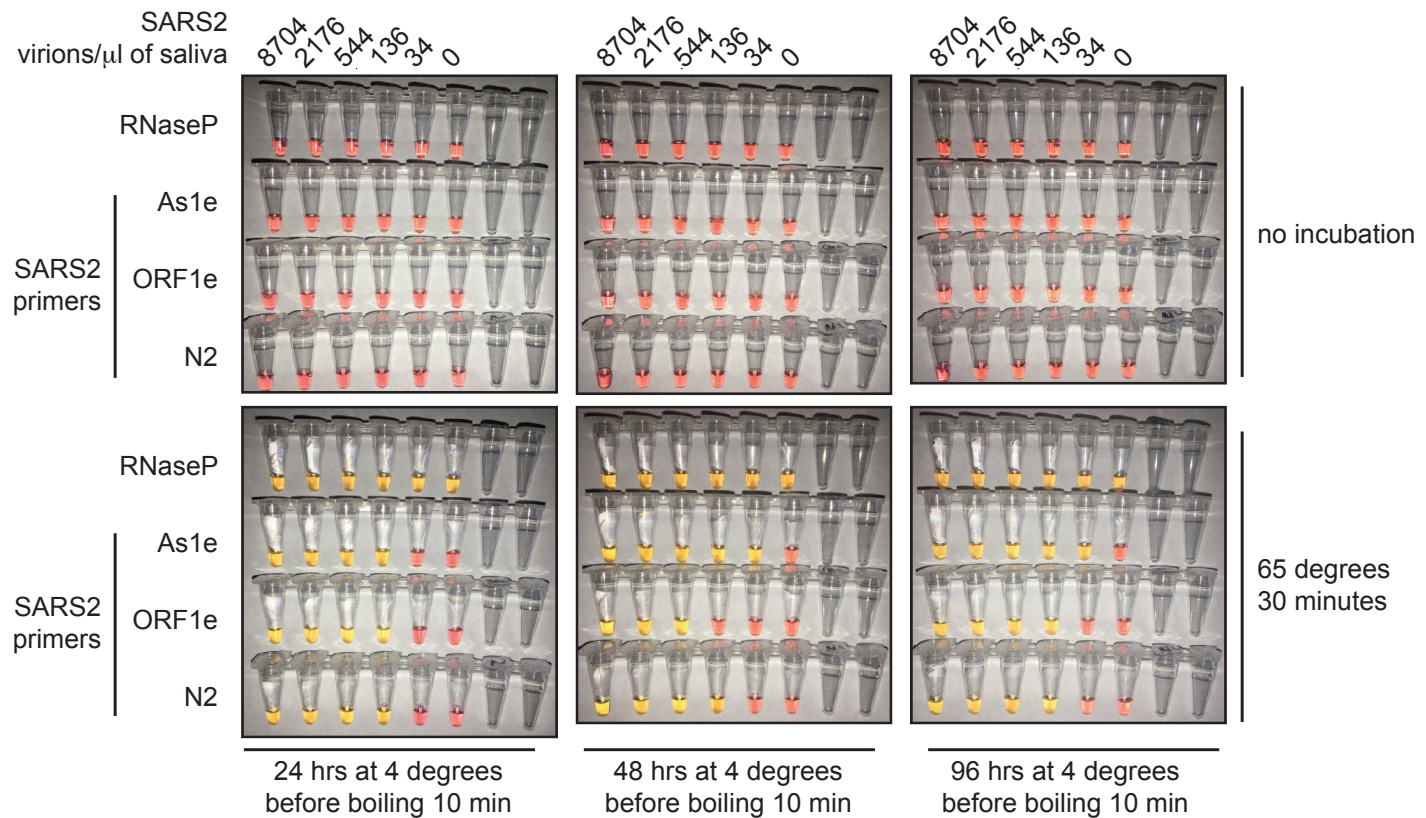


Figure S1: Optimized heat inactivation for detecting SARS-CoV-2 in human saliva. A) Saliva samples were diluted 1:1 with saliva stabilization solution and then spiked to reach the indicated concentrations of *in vitro* transcribed SARS-CoV-2 RNA in the saliva either before or after sample was heated at 95°C for 10 minutes, and then subjected to RT-LAMP at 65°C for 30 minutes. In this colorimetric version of RT-LAMP, reactions remain pink when no amplification occurred, and turned yellow if there was an amplification event. An RT-LAMP primer set targeting the human RNaseP transcript is included as a host RNA amplification control in addition to the three SARS-CoV-2 primer sets shown in panel A. **B)** Saliva samples were spiked with the indicated concentrations of heat-inactivated SARS-CoV-2 virions before being diluted 1:1 with saliva stabilization solution. Samples were then heated at 95°C for 10 minutes and subjected to RT-LAMP similarly to the experiment shown in panel B. **C)** Saliva samples were spiked with the indicated concentrations of heat-inactivated SARS-CoV-2 virions before being diluted 1:1 with saliva stabilization solution. Samples were then heated at 95°C for the indicated amount of time, and subjected to RT-LAMP similarly to the experiment shown in panel B. Without heating, no SARS-CoV-2 RNA can be detected with RT-LAMP, presumably because virions remain intact and the viral RNA is not accessible by the

amplification enzymes. Amplification is somewhat inconsistent at 5 and 30 minutes possibly because at 5 minutes hardly any RNA has been liberated, and by 30 minutes it has been largely degraded. However, 10 or 15 minutes at 95°C appears to provide just the right balance between liberating and preserving RNA. All reactions contain the AS1E primer set. Duplicates are presented at each time point.



Supplemental Figure S2: Saliva samples in saliva stabilization solution are stable at 4°C for at least 4 days before processing. Saliva samples were spiked with heat-inactivated SARS-CoV-2 virions at the indicated concentration and mixed 1:1 with saliva stabilization solution. Samples were stored at 4°C for either 24, 48, or 96 hours. Samples were then heated at 95°C for 10 minutes and analyzed using RT-LAMP with the indicated primer sets. Reactions were incubated at 65°C for 30 minutes.



Result interpretation

AS1E	N2	RNaseP	Result Interpretation	Report
+	+	+/-	SARS-CoV-2 detected	Positive SARS-CoV-2
If only one of the two targets is positive		+/-	Inconclusive result	Inconclusive
-	-	+	SARS-CoV-2 not detected	Not detected
-	-	-	Invalid Result	Invalid

Supplemental Figure S3 (continued)

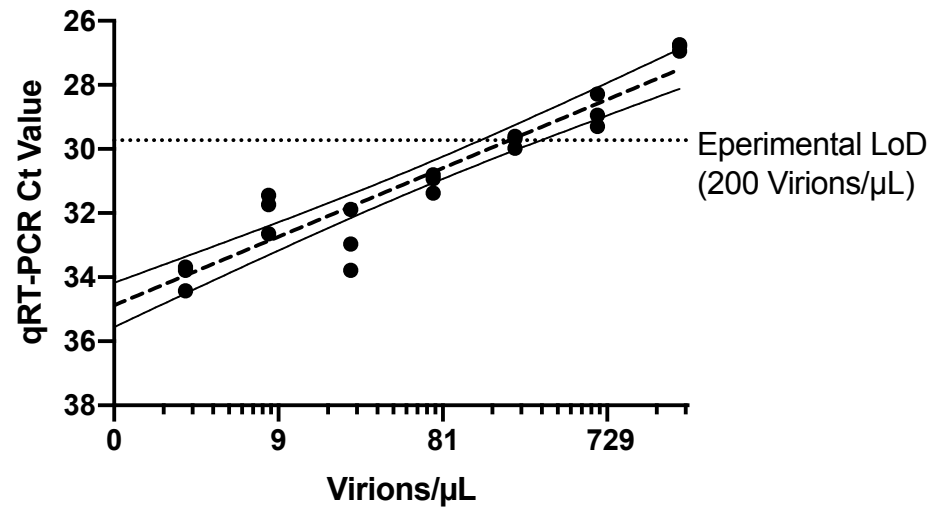
Summary of results from the contrived specimen study with saliva samples

	virions / μ L saliva	Number Tested	RNaseP	AS1E	N2	
1X LOD*	200	9	9 (100%)	9 (100%)	9 (100%)	Positive (%)
2X LOD	400	11	11 (100%)	11 (100%)	10 (91%)	
5X LOD	1000	5	5 (100%)	5 (100%)	5 (100%)	
10X LOD	2000	5	5 (100%)	5 (100%)	5 (100%)	
Negative	0	30	30 (100%)	0 (0%)	0 (0%)	
Total		60				

* 1X LOD (limit of detection) = 200 virions/ μ L

Summary of positive and negative agreement with contrived saliva samples

		Contrived Specimen Type		
		Positive	Negative	Total
Saliva RT-LAMP Test	Positive	29	0	29
	Inconclusive	1	0	1
	Negative	0	30	30
	Total	30	30	60
Positive Agreement		97% (29/30)		
Negative Agreement		100% (30/30)		



Supplemental Figure S4: qRT-PCR standard curve used to determine the Ct value to virion/μL calculation. 10,000 copies/uL of heat deactivated SARS-CoV-2 virus was spiked into negative saliva specimens from 6 different volunteers and incubated for 30 minutes at 95°C. Samples were diluted to indicated concentrations using heat-treated saliva without SARS-CoV-2 addition from the same individual. The standard curve generated from the linear regression analysis is illustrated with 95% confidence interval. The experimentally determined limit of detection for RT-LAMP is translated to quantitative RT-PCR Ct value of 30.4, as indicated by the horizontal dashed line.

Supplemental Text S1: Troubleshooting RT-LAMP False-Positives

See also Thi et al ¹⁵ for additional, excellent troubleshooting advice

Controlling acidic and variable human saliva samples: We found that the biggest obstacle to implementing the colorimetric RT-LAMP assay is the variability in the reaction pH condition. First, false-positive signal can result from saliva samples that are naturally acidic. We spent significant time addressing this issue and ultimately found that all samples must be over-buffered towards basic conditions in order to set the correct threshold for specificity in the test. This was achieved through a titration series of sodium hydroxide in the saliva stabilization solution to find the optimal concentration that would ensure all RT-LAMP reactions start pink and are still capable of turning yellow if amplification occurs. However, note that other components in the saliva stabilization solution (EDTA, pH=8.0, and TCEP-HCl) have also played a role in establishing the optimal pH. Additional saliva stabilization solution optimization might be needed if other forms of these components are used. Second, the colorimetric RT-LAMP reaction relies on phenol red to detect the pH change during the amplification. Any additional buffering agents, such as tris-acetate or tris-borate that are commonly present in laboratory reagents, should be avoided to prevent potential false negative signals, as these buffering agents tend to inhibit the pH change.

Controlling reaction acidification by carbonic acid: A second issue that has to be carefully controlled is the exposure of reaction components to the surrounding environment. When carbon dioxide from the atmosphere dissolves in water, it creates carbonic acid, which if present in high enough quantities can trigger the phenol red to turn yellow regardless of reaction state. Control measures should be implemented in three ways: First, we advise preparing the reaction mix (RT-LAMP master mix, primers, and water) right before sample loading. This is to prevent background amplification as well as the acidification of the reaction mix due to exposure to air. Additionally, dry ice should be avoided or completely isolated from the reaction components during sample transportation, as the exposure to the excessive carbon dioxide could also lead to acidification of the reaction mix. Second, during the 30-minute 65°C incubation, it is essential to completely seal off the reaction tubes to prevent vaporization of the reaction mix, as well as the infiltration of the water vapor if a water bath is used. We have noticed that an incomplete seal could lead to false positive signals. Last, because RT-LAMP amplification is highly robust, the test is very sensitive to contamination. Therefore, opening of reaction tubes after RT-LAMP has occurred should be strictly avoided as these tubes contain a large amount of target DNA. Alternatively, the NEB WarmStart LAMP 2X Master Mix with UDG (NEB M1804) can be used to eliminate DNA contamination. Through these results are not shown, we have verified the same limit of detection can be reached using this alternative master mix.

Controlling laboratory-based contamination: When carrying out the RT-LAMP SARS-CoV-2 screening test at scale, it is critical to assign isolated workstations, each containing their own set of laboratory equipment such as pipettes, centrifuges, vortexes, and cleaning supplies. This equipment should never move between stations and be regularly decontaminated using a detergent based cleaning solution such as 10% bleach or any other commercially available solution designed to eliminate nucleic acid contaminants. Additionally, special care should be taken by laboratory staff to regularly replace gloves if moving *backwards* from the following workstations:

Workstation 1: Setting up master mixes. This workstation is dedicated to making and aliquoting the master mix containing the RT-LAMP enzymes, primers, and water. All reagents should be centrifuged and spun down after thawing. As mentioned above,

master mix should be made and aliquoted shortly before addition of saliva (Workstation 2) to avoid carbon dioxide solubilization due to atmospheric exposure.

Workstation 2: Adding saliva samples to aliquoted RT-LAMP reaction tubes. This workstation is dedicated to handling the processed saliva samples. Once saliva samples are added to the aliquoted RT-LAMP reaction tubes, care should be taken to ensure that an appropriate seal is established (e.g. dome cap strips) to minimize airflow during reaction incubation. This workstation should include two micropipettes capable of pipetting a volume of 2 µl. One pipette can be used for saliva samples, while the other pipette should be used exclusively for pipetting any RNA controls (e.g. *in vitro* transcribed SARS-CoV-2 RNA).

Workstation 3: RT-LAMP reaction incubation and results reporting. This workstation contains the heating element (e.g. heat block, thermal cycler) where RT-LAMP reactions are incubated. This Workstation has the highest risk of contamination, since the RT-LAMP reaction products will be in high abundance and can themselves serve as a template in subsequent reactions. When reactions are removed from the heating element, they should immediately be analyzed, results recorded, and then reaction tubes should be disposed in a container with a lid. *Never carry completed reaction tubes to any other part of the lab. Never open completed reaction tubes for any reason.* Any laboratory technician that has entered Workstation 3 should dispose of their gloves before returning to any other part of the lab.

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